

This is sample data provided for CALIBER purposes only		ONE POINT LESSON (OPL)		OPL No: OPL-EA-5601-05 Sheet 1 of 1
OPL Title	Tube Leak Detection & Plugging	Discipline	Mechanical	
Equipment	EA-5601 - SOLVENT HEATER	Area / Unit	5600 - SOLVENT HEATING SYSTEM	
Related Interlock	N/A (control loop only) (N/A)	P&ID Ref	TJC-LLD-PID-5601	
Classification:	☐ Basic Knowledge	☐ Improvement	☒ Trouble Case	Aspect: P-Q-C-D- S -M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **tube leak detection & plugging** for EA-5601 (SOLVENT HEATER) in the SOLVENT HEATING SYSTEM. A tube leak lets steam into the solvent (rising outlet temperature and water in the solvent); the leaking tube must be found and plugged. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **LOW CRITICAL** handling **Hexane / MP Steam** at up to 16 barg (tube) / 200 degC. Observe the following before and during the task:

- Apply LOTO (Lock-Out Tag-Out) and prove zero energy before starting work.
- Isolate, depressurise, drain and purge the equipment; obtain a valid work permit.
- Wear the correct PPE: coverall, safety shoes, gloves, goggles and hearing protection.
- Beware of stored mechanical energy (springs, rotating inertia) and pinch points.
- Use certified lifting gear and correct manual-handling technique for heavy parts.

3. TOOLS & MATERIALS REQUIRED

- Standard mechanical hand tools & calibrated torque wrench
- LOTO padlocks and permit
- Laser/dial alignment kit (as applicable)
- Feeler gauges, dial indicator, seal/bearing pullers
- Correct spare parts and gaskets per the BOM
- Lifting gear and slings (certified)

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Isolate and hydrotest the shell side to find the leak	Isolation & LOTO verified
2	Bubble-test to identify the specific leaking tube	Parts match BOM / spec
3	Plug both ends of the tube with tapered plugs	Torque / clearance to spec
4	Record the plugged-tube count (maximum 5 % of tubes)	No leak / smooth operation
5	Plan bundle replacement if the plugging limit is reached	Record in equipment history

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
High tube dP PDT-5605, low solvent outlet temperature	Hexane fouling / polymer fines deposit on tube bores	Hydrojetted 246 tubes, restored dP and duty, re-gasketed
Hexane weep at channel flange	Spiral-wound channel gasket relaxed / damaged	Renewed channel gasket, re-torqued in cross pattern
Continuous steam at condensate, high steam use	Steam trap element failed open	Replaced steam trap element, verified condensate temperature

6. KEY LEARNING POINTS

- A tube leak lets steam into the solvent (rising outlet temperature and water in the solvent); the leaking tube must be found and plugged.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-5601 before working on EA-5601.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
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Cross-ref tag GA-1201A across Datasheet, GA, P&ID, Interlock and Maintenance History.